

Using discussion forums to explore attitudes toward cars and licensing among young Australians



Alexa Delbosc*, Graham Currie¹

Public Transport Research Group, Institute of Transport Studies, Department of Civil Engineering, Monash University, Building 60, Clayton, Victoria 3800, Australia

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ABSTRACT

Young people are becoming increasingly less likely to acquire a car license in North America, Australia and much of Europe. Emerging research is beginning to explore the structural and demographic explanations for these trends. However little research explains how attitudes may be shaping the decisions young people make about cars and driving. Popular hypotheses are the (i) changing social status of the car, (ii) role of electronic communications and (iii) environmental awareness, but little academic research has directly explored these issues.

This paper uses innovative online research tools to explore these issues with young people. Three discussion forums were run with participants from across the state of Victoria, Australia. 33 People aged 17–23 contributed across the course of one week's discussion.

Discussions suggest that among young Australians the car itself may not be a symbol of status and luxury, but instead a symbol of adulthood and maturity. Electronic communications were seen as a supplement to face-to-face contact, not a replacement for car travel. Not one person in the sample spontaneously mentioned that environmental concerns shaped their travel choices; even when prompted these concerns were far removed from travel decisions.

The paper follows with a commentary on the use of discussion forums for qualitative research. It closes with a discussion of the new hypotheses that these findings generate and uncovers many areas for future research.

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1. Introduction

Reliance on the car has serious negative consequences for society. On-road transport contributes 17% of global carbon dioxide emissions (Chapman, 2007) and the World Health Organization estimates that globally more than 1 million people are killed and 20–50 million are injured by road collisions each year (Nantulya et al., 2003). Against this backdrop there is a remarkable emerging trend which is not well understood: young people in much of the developed world are less likely to initiate car-driving, and if they can drive, they are driving less (Raimond and Milthorpe, 2010; Kuhnimhof et al., 2011; Sivak and Schoettle, 2012a, 2012b).

There is a large body of research on understanding mode choice and car ownership more generally (e.g. Scheiner and Holz-Rau, 2007). However the downward trend in car travel amongst the young has only recently been recognised (Delbosc and Currie, 2013). Researchers, government bodies and even the car industry

are beginning to question why this is occurring. Some initial research has focused on economic and socio-demographic influences on car licensing (e.g. McDonald and Trowbridge, 2009; Raimond and Milthorpe, 2010; Kuhnimhof et al., 2012b; Delbosc and Currie, in press). Three key attitude shifts are also hypothesised but lack a strong empirical basis: (i) the changing social status of the car, (ii) the growing role of electronic communications and (iii) a growing environmental awareness amongst young people.

This paper uses innovative online recruitment and research tools to explore how young people talk about transport and travel as a means of providing input to this debate. It reports on qualitative discussions conducted with young people in Victoria, Australia, to better understand their thoughts and feelings about car travel, getting a license and owning a car with a focus on the three hypothesised attitudes noted above. The purpose of the research is to illustrate broad themes that can be explored in further research. The paper also provides an assessment of the new online research tools adopted.

The paper takes the following structure; the next section contains a review of past literature on young people and travel, focusing on the three proposed hypotheses for reduced use of cars

* Corresponding author. Tel.: +61 3 9905 5568; fax: +61 3 9905 4944.

E-mail addresses: alexa.delbosc@monash.edu (A. Delbosc), graham.currie@monash.edu (G. Currie).

¹ Tel.: +61 3 9905 5574; fax: +61 3 9905 4944.

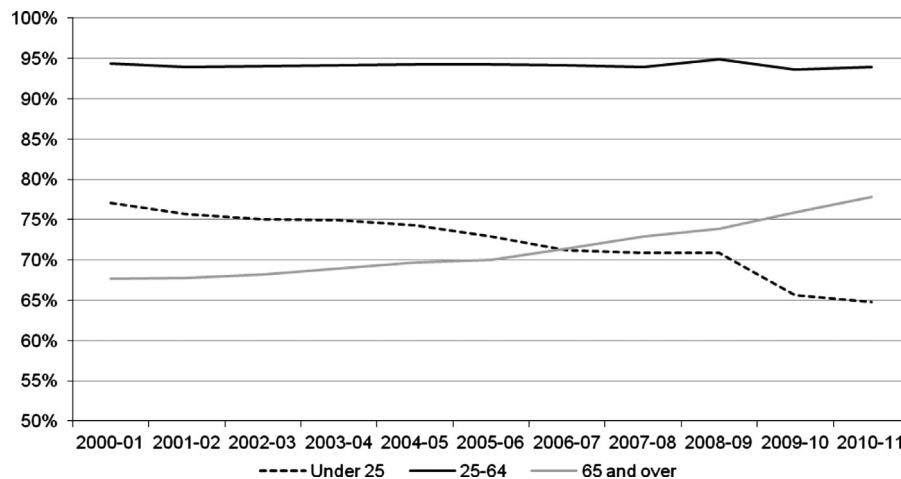


Fig. 1. Licensed drivers as percent of age group, 2000–2011, Victoria. Note: data sourced by Australian financial year (July–June). Source: licensing data supplied by VicRoads (VicRoads, 2012) and compared to population (Australian Bureau of Statistics, 2010).

by young people (items i–iii noted above). This is followed by a description of the research method and the online research tools adopted. The results of the discussions are then presented. A short reflection on effectiveness of the online research tools follows. The paper ends with a discussion and conclusion highlighting areas for future research.

2. Research context

In much of Europe, North America and Australia, licensing rates, automobile ownership and use are steadily decreasing (Kuhnimhof et al., 2012a; Sivak and Schoettle, 2012a, 2012b; Blumenberg et al., 2013; Delbosc and Currie, 2013; Smart, 2013). For example, amongst young people in the Australian State of Victoria, the percent of under-25 year olds with a driver's license dropped from 77% in the 2000–2001 financial year to 65% in 2010–2011 (see Fig. 1) (Australian Bureau of Statistics, 2006; VicRoads, 2012). Similarly in the United States, licensing of 20–24 year-olds has dropped from 92% in 1983 to 81% in 2010 (Sivak and Schoettle, 2012b).

Emerging research suggests that situational variables such as lower household incomes, residential location, parental lift-giving and household structural changes (such as a decrease in full-time youth employment and postponement of marriage and child-rearing) are contributing to these trends (McDonald and Trowbridge, 2009; Raimond and Milthorpe, 2010; Kuhnimhof et al., 2012b; Delbosc and Currie, 2013, in press). The impact of graduated license schemes, which have been introduced in many countries, is also highlighted as a possible reason (Raimond and Milthorpe, 2010) although thus far it has only been found to have a minor impact (Smart, 2013).

However in addition to these structural variables, a number of researchers have begun to consider whether the attitudes of young people have contributed to this decline in car reliance. Three overlapping attitudes have been identified by researchers and the popular media: the status of the car, the rise of electronic communications and the environmental concerns. Much of the following research is drawn from the grey literature of consulting and market research reports, which means methodological details are sometimes lacking.

2.1. Is the social status of the car changing?

Many social researchers have studied the role of the car as a status symbol, a symbol of prestige or an object of desire (Stokes

and Hallett, 1992; Hiscock et al., 2002; Steg, 2005; Redshaw, 2006). There is an emerging sense that these values are shifting amongst the younger generation as interests shift from cars to computers and electronic gadgets although research is almost exclusively cross-sectional, not longitudinal. For example a consultancy survey conducted in 2008 in Japan found that 27% of 40–59 years-old listed cars as an interest, overall ranking 7th (just ahead of computers). This was only 23% for 18–24 years old university students, ranking 17th behind such interests as personal computers, portable music players, communication devices, anime, video games and TV (Kalmbach et al., 2011).

This alleged trend is capturing much attention in popular press in Australia, North America and Germany (Kuhnimhof et al., 2012c) with newspaper headlines like 'Young people choose computers over cars' (Campbell, 2012). Car manufacturers are also concerned about young people losing interest in their products (Chozick, 2012).

However thus far much of this concern is speculative, based on the popular press rather than academic research. The academic literature has little to say on the topic and not all of it is clear-cut. For example, at least one study of Dutch people with car licenses found that younger respondents were slightly more likely than older generations to value their car for symbolic reasons (e.g. 'my car gives me prestige') and affective reasons (e.g. 'I love driving') (Steg, 2005). There is clearly disagreement within academic literature over whether young people are less likely to view cars as a status symbol.

2.2. Are electronic communications reducing the need for car travel?

The idea that the younger generations prioritise gadgets over cars is coupled with the belief that electronic communications (such as Facebook, texting or online chat) are reducing the need for young people to see their friends in person. This suspicion is somewhat supported by an academic survey in the United States that found that a small portion (5–10%) of 15–18 years olds had not started the licensing process in part because 'Facebook, texting etc. keeps me in touch with friends' (Williams, 2011); however no respondents cited this as their primary reason. A commercial survey sponsored by the car-sharing scheme Zipcar found that over 50% of young people agreed that they sometimes chose to spend time with friends online instead of driving to see them (KRC Research, 2010), although methodological details about this survey are scarce.

A recent analysis by Sivak and Schoettle (2012a) found that countries with more internet users had lower overall licensing rates. The popular press has quickly leapt on this study as evidence that young people are using the internet to replace face-to-face contact: 'Today Facebook, Twitter and text messaging allow teenagers and 20-somethings to connect without wheels' (Chozick, 2012, p. 1). Yet this conclusion is premature because the study measured household internet penetration, not internet use by young people; the validity of this evidence is currently being debated in the academic literature (Le Vine et al., 2013; Sivak and Schoettle, 2013).

Another shortcoming of this early work is that it rarely makes the distinction between reducing the need to travel and reducing the need to drive. If e-communications truly reduce the need for face-to-face contact, this should also reduce the need for travel by all modes, not just by car, which is still an open question. This clearly highlights the need to better understand the role that electronic communications play in maintaining social contacts amongst young people.

2.3. Are environmental attitudes changing?

Media reports about the declining status of the car also cite environmental awareness as a related contributing factor. For example a commercial survey commissioned by the car-sharing company Zipcar found that almost half of young people agreed with the statement 'I want to protect the environment, so I drive less' (KRC Research, 2010).

However academic research casts doubt over whether young people are actually more environmentally aware than older generations, and more importantly, whether environmental attitudes actually influence their behaviour. One longitudinal survey in Sweden between 2002 and 2009 did find a slight increase in 17-years-olds citing environmental concerns as a reason not to get a license (Forward et al., 2010). However a different survey in Sweden found that adolescents were no more likely than their parents to believe that cars have a negative impact on the global environment (Sandqvist and Kriström, 2001). Furthermore a cross-sectional British survey found that compared to all other age groups, people aged 16–24 were the least likely to say they were 'environmentally friendly in most everything they do' and were the most likely to say they do not do anything environmentally friendly (Department for Environment Food and Rural Affairs, 2009). These findings cast serious doubt over whether changing youth car reliance can be credited to a change in environmental attitudes.

Overall, this literature review highlights the paucity of evidence about the influence of social status, e-communications and environmental attitudes on youth driver licensing. Much of the existing research comes from commercial and market research which lacks transparency and peer review. Even the existing academic literature does not paint a clear picture and controversies are being actively debated (e.g. Le Vine et al., 2013; Sivak and Schoettle, 2013). This paper provides new insight into these issues using qualitative research methods.

3. Research method

The increasing uptake of home internet connections has seen a similarly rapid development of online research methods. Web surveys are now a common method of quantitative data collection; however qualitative research has been slower to explore online methods (Mann and Stewart, 2000). Since the turn of the millennium, qualitative online methods have been applied in health, market research and sociology and are considered particularly

appropriate where discussion participants are geographically dispersed or the subject area is personally sensitive (e.g. body image) (Fox et al., 2007). They are also considered particularly appropriate when studying young people, who may have greater difficulty accessing conventional focus groups (especially if they do not yet have a driving license) and are considered particularly competent in communicating online (Fox et al., 2007).

Online focus groups may be held in real-time (synchronous) through chat rooms or non-real-time (asynchronous) through discussion forums (Mann and Stewart, 2000). An asynchronous discussion forum was selected for this study as it allowed young people the freedom to participate in their own time and also to consider responses to the questions raised.

3.1. Recruitment

Recruitment was carried out in two waves. In the first wave, 4891 people from an online research panel were contacted by a market research company, 69 (1.4%) filled in a screening questionnaire, qualified for the discussions and opted-in to be contacted with further details. Of these, 28 people completed the registration process to participate in the discussion giving an overall response rate of 0.6% from initial contact.² Four of these respondents had very low participation rates (between zero and five posts), although their responses were included.

This first wave of recruitment only recruited two people without a license. Consequently, a second recruitment company was employed to recruit non-licensed persons using traditional telephone recruitment. 14 People agreed to participate and of those five people completed the registration process and contributed to the forum.

In both waves of recruitment, participants were told that the discussions would be about the general topic of 'drivers licensing and travel amongst young people'; they were not told that licensing rates are declining. More specific details about the discussion topics were not introduced until participants had registered (see Section 3.2).

Three discussion groups were run. One contained 13 respondents from the greater Melbourne region. The second group consisted of 15 residents from a range of towns and remote areas across Victoria. The third group of five unlicensed persons was from Melbourne. As Fig. 2 demonstrates, their home locations are geographically diverse which would be difficult to attract using conventional recruitment methods and face-to-face delivery of focus groups.

It should be emphasised that the purpose of qualitative research is not to gain a representative sample but to illustrate broad themes that can be explored in further research. However Table 1 gives an overview of general characteristics of participants compared to the population of Victoria. The discussion groups had a higher proportion of female participants and people from outside of Melbourne. Around half of participants had a P-plate or full license that allowed them to drive without supervision; this is slightly lower than the state average. Most participants lived at home with their parents and were studying, at slightly higher rates than the state average. In Section 4 of this paper, participants will be identified using a code representing their gender (M or F), age and license status (N, L, P or F – No License, Learner, Provisional or Full License respectively).

² We acknowledge that this is a very low response rate, but this is because we initially approached a company that specialises in online questionnaires. Their respondent pool was not used to being recruited for qualitative research, which is why a traditional focus group recruitment company was used for the second wave of recruitment.

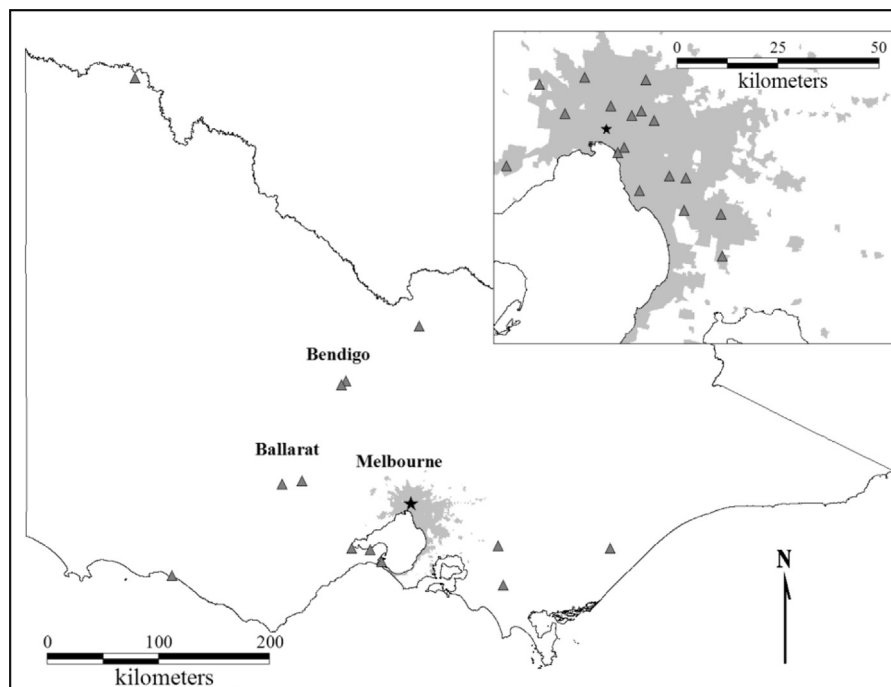


Fig. 2. Metropolitan and regional focus group participant home locations. Note: star represents Melbourne's central business district.

Table 1
Characteristics of focus group participants

Characteristic	Sample (%)	2011 Census ^a (%)
Location		
Regional Victoria	45	22
Metropolitan Melbourne	55	78
Gender		
Female	64	49
Male	36	51
License status		
None	24	n/a
L-plate ^b	27	n/a
P-plate ^c	42	n/a
Full licence	6	n/a
Total P-plate or full license	48	65 ^d
Household makeup		
Living with parents	73	60
Living with roommates	12	10
Living with spouse/partner	12	7
Living alone	0	3
Employment status		
Studying	61	56
Working part time	18	n/a
Working full time	6	n/a
Unemployed	12	n/a
Home duties	3	n/a
Age		
17	12	14
18	21	14
19	12	14
20	21	14
21	21	15
22	6	15
23	6	15

^a 2011 Census data for those aged 17–23 for the state of Victoria (Australian Bureau of Statistics, 2011). Employment data not available at time of publication.

^b Learner's permit: can only drive under supervision.

^c Provisional license: can drive independently with restrictions on blood alcohol and number of passengers.

^d Data supplied by VicRoads, the state licensing authority.

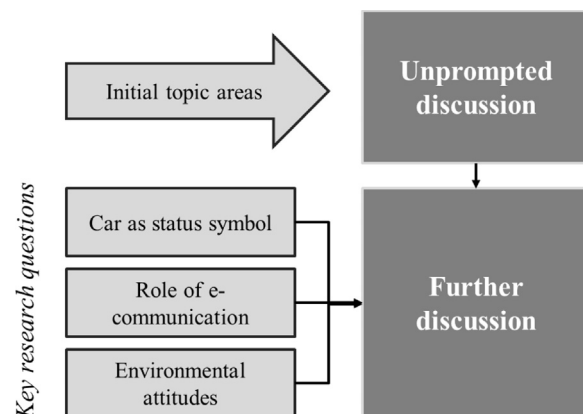


Fig. 3. Discussion flow.

3.2. Approach

The discussion forum was hosted by the Monash University using the program 'Sakai' (<http://www.sakaiproject.org/>). Six discussion topics were set up to stimulate discussion and further topics were introduced after a few days. Participants were required to post at least twice on every topic to encourage repeated participation but they were able to post at any time that suited them. The forums were open for 1 week and participants who met the required number of posts were offered a gift voucher. Researchers checked the forums throughout the week to ask questions, stimulate further discussion, and check for inappropriate or abusive posts (no abusive posts occurred).

The three primary areas of research interest were the three hypotheses: (i) cars as status symbols, (ii) the role of electronic communication and (iii) attitudes toward the environment. However these three topics were not introduced at the start of

discussions in order to gain an understanding of how ‘top of mind’ these issues are for participants, although it was noted where these topics came up unprompted (see Fig. 3). After a few days, these topics were gradually introduced to gain a more direct understanding of the topics.

Instead, six general discussion topics were presented at the beginning of the week:

1. Describe your travel in a typical week. Where do you go and how do you usually get there? How do you choose what travel mode to take (e.g., drive, walk, and public transport)? How well do your current travel arrangements work for you?
2. In your opinion, what sort of city is Melbourne [‘your local area’ for the regional group] in terms of transport and travel? What sort of city would you like it to become? Would those changes actually change anything in your life – like where you’d want to live or how you’d travel?
3. What do you think of the process of getting an L-plate and P-plate license? If you don’t have a license, are you planning on getting one in the near future?
4. How important is owning a car to you? If you don’t have your own car, what would it mean to you to have one? Do your friends see cars the same way you do?
5. Who pays for the car, petrol, insurance, etc.? How important are these costs to you? If you don’t have a car, is cost a major factor in this?
6. Will how you travel today change over the next 5, 10 or 20 years? What would have to happen to change the way you travel today?

These original topics were adjusted to suit the groups (for example the unlicensed group was not asked about paying for car expenses) and they were supplemented by follow up questions. Not all of these six topics will be discussed in this paper.

Responses were analysed systematically by first identifying how (if at all) the three topics were discussed in unprompted discussion. In the prompted discussion section, responses were integrated to identify key themes and areas of difference and disagreement. Quotations were selected where they best illustrated key points.

4. Results

The initial six topics provided a wealth of discussion around young people’s travel, more than can be reported in this paper. However a few contextual points are worth mentioning.

Travel habits varied greatly. Some were captive public transport users with very complex, multi-modal travel involving walking, lifts and multiple transit modes. Some expressed an explicit preference for public transit over car travel, even showing a remarkable level of tolerance for infrequent services, long walking distances or long travel times on public transport.

Others expressed quite entrenched dependence on their car. Across all discussion topics, in both the metropolitan and regional groups, cars were described with reference to providing freedom and control over their time and activities. In the words of one participant, moving from a reliance on lifts and shared family car to owning his own car made ‘all those problems disappear’ (M21P). They acknowledged that even though a car was a financial burden, this cost was worth it to ‘be able to drive wherever you like when you want to’ (M18P). This dependence was heightened in regional and rural areas where one young woman spoke of being lost without her car and having to ‘resort to begging lifts’ when her car was being repaired (F21P).

In the first few days of discussion, the three key research areas were not directly addressed by the researcher. This was an explicit choice to determine whether any of the three areas were ‘front of mind’ for young people when they talked about travel. After a few days, the three proposed hypotheses were explored to different parts of the discussion to bring the topics into the explicit discussion. The following sections describe how this was achieved and the key outcomes derived.

4.1. The car and social status

4.1.1. Unprompted discussion

The symbolic social status of a car can be a difficult thing to directly measure. There were very few spontaneous expressions of social status around whether or not someone had a car. Even in the regional discussion where car licensing and ownership were considered a ‘given’ and an ‘expectation’, this was expressed in practical terms as a necessity ‘that you probably can’t live without’ (M20P).

Instead, cars were usually discussed as a responsibility and a symbol of adulthood and freedom. It is referred to as an achievement, a part of growing up and an expression of independence.

A car to me symbolises commitment, financial responsibility and to some extent becoming an adult/losing one of the last parts of my childhood. (F22N)

There were a variety of arrangements across participants including shared ‘family’ cars, parental help with payments and full purchase and payment of vehicle costs. Those who purchased their own car sometimes express a sense of jealousy toward those who have a car purchased for them. Furthermore, expressing this sense of responsibility and maturity implies that people who do not pay their own expenses have not reached this level of maturity. Being keenly aware of expenses can generate frustration with those who do not have to face these responsibilities:

It bugs me when my friends who don’t drive show no regard for the expenses of driving, and want me to drive them around without realising how expensive it can be. They rarely offer to chip in for petrol, and I don’t expect it, I just get frustrated that it’s something that doesn’t even occur to them because they don’t have to worry about it. (F21F)

Many participants did talk about the love they had for their car, suggesting that cars do capture the interest and affection of many young people. However this affection was not couched in regards to social status and was often expressed toward old or decrepit cars: ‘I love my car because I have built it from the ground up, even though it is not one of the best cars around it means a lot to me’ (M21P).

4.1.2. Prompted discussion

Several days into the discussion group the moderator specifically asked whether participants (or their friends) saw a car as a status symbol. The presence of a car did not appear to infer status in and of itself, but participants did state that owning an expensive car conferred higher social status than an old or inexpensive car. They also mentioned building or modifying cars as a reflection of pride in the vehicle. One young woman with a learner license mentioned that she was not jealous of her friends’ cars, but instead was jealous of ‘trivial’ things that concerned her like ‘the latest gadget’ (F19L).

One of the unlicensed participants did mention that he would feel ‘silly’ if he still did not have a license by the time he was 30. However he saw having a license as a basic skill rather than a status symbol:

Most of my friends do not have cars, but we are all still fairly young (early 20 s). Once you're dealing with someone in their 30 s who can't drive, they appear to me like someone without a very basic, foundational skill, that all people should have. It would be like being illiterate - it's just assumed that everyone over a certain age can read/drive... Also, guys in suits on public transport, complete with backpacks, do look childish. I wouldn't want to look like that. (M22N)

Although qualitative research is not meant to provide a representative sample of all young people, these discussions did not uncover a strong sense of the car as a status symbol but rather as a symbol of adulthood.

4.2. Travel and the role of electronic communication

4.2.1. Unprompted discussion

Much of the early discussion of travel was couched in the context of socialising with friends and maintaining friendships, especially for the younger participants. In the words of one participant, seeing friends whenever he wanted was 'the most important thing in the world' (M18P). There was a clear contrast present when some members of a friendship group had access to a car and others did not. Those without access to a car spoke of having to 'mooch' (persuade others to provide) lifts off people and being unable to meet up when they are 'randomly' invited to hang out. One young woman with a car expressed frustration with her closest friend who does not have a license: 'it gets very annoying having to always pick her up, drop her off, drive her places, have her turn up late for everything coz she missed her bus, etc.' (F21F).

In addition, relying on public transport or lifts to socialise was sometimes spoken of with frustration. Some of those without a driving license spoke of public transport as a barrier to maintaining friendships. They spoke of 'drifting away' from friends and the frustration of buses and trains not running late at night when they wanted to socialise.

Importantly, all of the initial discussion of socialising focused on how people visited their friends in person. Not one person spontaneously suggested that electronic communications replaced or reduced the need to see friends, by any mode of travel.

4.2.2. Prompted discussion

Several days into the discussion, participants were specifically asked 'Do you think Facebook, mobile phones and IM [instant messaging] chat can make up for not seeing your friends in person?' In the resulting discussion, it was clear that electronic communications were seen as a complement to face-to-face contact with friends, not a substitute. Online communications were seen as a way to organise times to meet up in person or a way to 'keep in touch' with past friends. Electronic-only bonds were described as somewhat tenuous; one young woman states that 'all the friends that I just talk to online are slowly drifting away' (F19L). However some young people with restricted car access found themselves in a position where electronic contact became their only contact:

As I do not drive - I do find it difficult to maintain friendships. Especially now that I have entered university - it is absolutely impossible to not drift away from previous high school friends. Social network sites, mobiles and IM chat is basically the handy way these days to remind my high school friends that I am still around and is essentially what strengthens our relationships despite its virtual form... However it does not replace the essential bonding through face to face contact, and I do feel that because I do not have my own transport - my bonds with previous friends are not as strong as it [sic] used to be (F19L).

These findings cast some doubt on the hypothesis (e.g. [Sivak and Schoettle, 2012a](#)) that e-communications are reducing the need to travel by car. However they may be a 'satisficing' method that can help maintain friendships that would otherwise have been lost without e-communications.

4.3. Environmental attitudes and travel

4.3.1. Unprompted discussion

When the discussion groups were opened, none of the topic prompts mentioned the environment. This was a conscious decision in order to see whether young people spontaneously mentioned environmental concerns in the course of the discussion. There were very few spontaneous comments about the environment in initial discussions. No participants specifically mentioned environmental concerns as a factor influencing their mode choice or decision to get a driving license.

4.3.2. Prompted discussion

After several days a new discussion topic was added asking young people to reflect on the environment, health and safety impacts of their travel choices. Even after this prompt, quite a few people chose to discuss health and safety and did not even discuss environmental impacts. Of the non-drivers, none said that environmental concerns were a major reason they did not drive, although it did appear to make them feel better about not having a license.

Instead, many people expressed a degree of ambivalence around environmental concerns. Many said they were conscious of the environment and expressed a desire to travel sustainably. But this desire was usually immediately followed by justifications for why they could not do anything about it.

I do consider the environmental impact my travel is having, however I feel as though there's not a lot I can do to help the situation when it comes to travel. Like others, I cannot walk/cycle to work/uni etc., and my car is not a gas-guzzler, nor does it pump out fumes. (F21F)

Many young people expressed a sense that their impacts were minimal and there was 'nothing they could do'. However others discussed strategies they used to reduce their impact, such as carpooling or trip chaining. One young woman said that whilst she could not do anything about her travel she instead made sustainable choices in other aspects of her life.

Some young people outright rejected the idea that their travel choices could have a negative impact at all. One young man, for example, assumed that because his car did not emit visible smoke fumes like 'those massive diesel trucks' then his impact was negligible (M21P). Another expressed doubt in climate change altogether: 'I honestly can't say I have really thought about it at all. It just hasn't been a prominent thing in my mind as I'm not a big believer in climate change.' (M18P)

This sense that 'nothing could be done' in the present was sometimes coupled with an expression that things would be different in the future. Some young people were confident that petrol-based engines were on their way out and would soon be replaced by alternative fuels. Yet these alternatives were described in a more abstract, future-oriented way. Perhaps this future focus is used by some as a justification for why they did not have to change their travel habits in the present.

5. Reflections on discussion forum method

This section provides a short summary of the strengths and weaknesses of using online focus groups to collect qualitative data.

There were two main disadvantages to the method: recruitment difficulties and difficulty in guiding the discussion. Although online methods are meant to make it easier to organise focus groups, some difficulties were experienced in finding a recruitment company that could meet the needs of the project. Recruitment rates from the online pool were very low as this recruitment company specialised in online questionnaires, not qualitative research. This is why a traditional focus group recruitment company was employed to recruit participants without a license. This reflects not so much on the discussion forum format as on the difference between online and telephone recruitment, and it is strongly recommended that telephone recruitment be employed with this method.

Guiding the discussion also proved somewhat difficult. Although the discussions generated meaningful responses, the asynchronous format made it difficult to probe for greater depth of responses. Several times the moderator asked someone to expand on or clarify a point but the participant either did not notice the question or did not want to respond. A 'new reply' notification system, perhaps through email notifications, may help resolve this issue. Furthermore after using this method with another research topic it was found that discussions were richer when respondents were told they must reply once a day for 3 consecutive days, as opposed to whenever they want over the course of a week.

Overall, however, the online focus groups were considered a qualified success. The primary strength of the method was the ability to engage with a geographically dispersed and difficult-to-contact group of young people scattered across Melbourne and the state of Victoria. The method also kept costs down, although this will vary depending on how participants are recruited. Traditional focus group recruitment methods can be relatively expensive per participant although this will depend on the company employed.

Furthermore, participants appeared to enjoy the discussion. Several young people told the moderator they found the discussions interesting and that they learned a lot about what other people think about transport issues.

6. Discussion and implications

This paper uses online research tools to explore three common attitudes of young adults: (i) the social status of the car, (ii) the role of electronic communications and (iii) environmental awareness. These discussions provided some early insights into these three attitudes and highlighted some important areas for future research.

Although there was some discussion of expensive cars conveying higher social status, the basic act of owning a car was seen not as a symbol of status but rather of 'grown up' responsibility. People who did not own a car were not characterised as lower status; rather they were sometimes spoken of as more dependent or less mature. If cars are now seen as a step toward adulthood, this is likely to interact with recent findings that a decline in licensing is associated with young people delaying life stages such as moving away from home, getting a full-time job and having children (Delbosc and Currie, *in press*).

Without directly-comparable, longitudinal data it is difficult to conclude whether these attitudes are changing, although considerable research suggests that older generations saw the car as a symbol of status (Stokes and Hallett, 1992; Hiscock et al., 2002; Steg, 2005). Future research could use focus groups of different age cohorts or attitudinal surveys to compare the perceptions of different age groups toward the car as a status symbol. Even an ethnographic study of popular culture, such as movies and songs, could reveal some interesting insights into whether the portrayal of cars has significantly changed since the middle of the 20th century.

Media and grey literature comments on the decline in young adult driving quite often hypothesise that e-communications replace the need to drive (Campbell, 2012; Sivak and Schoettle, 2012a). In these discussions, e-communications were seen as a way to facilitate and strengthen face-to-face friendships; they were not seen as a substitute for travel in general, let alone driving specifically. Furthermore, media coverage often simplifies the complex role that e-communications can play in shaping travel. For example, if young people are feeling an increasing need for constant electronic connection, then perhaps this will influence some to travel by transit (where one can text or check email) over car driving (where texting is generally illegal). Or perhaps young people are more likely to prioritise paying for mobile phones or iPads over paying for petrol and car insurance, such as the young woman who was jealous of her friends' 'latest gadgets' but not their cars (F19L). Clearly this is a complex, yet important, area for future research; related research on e-commerce highlights how evolving technologies can influence travel habits (Golob and Regan, 2001).

Not one person in the sample spontaneously mentioned that environmental concerns shaped their travel choices. Even when an environmental topic was introduced by the moderator, reactions were quite mixed. Even non-drivers did not state that environmental concerns were a major factor in their choice not to acquire a license. Instead, car drivers justified why they 'had no choice' but to use their car. These findings provide no support for the view that the environment is a causal factor in changes in car travel behaviour of young adults. This is particularly telling when considered next to surveys suggesting that the younger generation is less concerned about the environment than older generations (Department for Environment Food and Rural Affairs, 2009).

It is important to note that qualitative research is not meant to stand in for a representative, longitudinal sample of views. There are, for example, significant sub-cultures of youths who place a great deal of social status on owning, modifying, and showing off their cars. Furthermore cross-sectional research cannot on its own pin down how these attitudes may have changed over time. Expanding the research to measure the attitudes of older generations is one method that may provide a broader understanding of attitude change.

This study is the first step in what is becoming a growing body of research on young people and travel choices (e.g. Delbosc and Currie, 2013). Future research will complement these qualitative findings with quantitative measures of attitudes and behaviours alongside situational influences such as income, living arrangements, work status and land use so that findings can be representative of wider trends. Only then can the relative influence of attitudes be directly compared to situational variables in order to give a better understanding of why young people are becoming increasingly less car reliant into the future.

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References

- Australian Bureau of Statistics, 2006. Census of Population and Housing.
- Australian Bureau of Statistics, 2010. Population by Age and Sex. Victoria. 3101.0.
- Australian Bureau of Statistics, 2011. Census Table Builder.
- Blumenberg, E., Wander, M., Taylor, B.D., Smart, M., 2013. The times, are they a-changin'? Youth, travel mode and the journey to work. In: Proceedings of the Transportation Research Board 92nd Annual Meeting, Washington, DC.

- Campbell, M., 2012. Young People Choose Computers Over Cars. The Age, Melbourne.
- Chapman, L., 2007. Transport and climate change: a review. *J. Transp. Geogr.* 15 (5), 354–367.
- Chozick, A., 2012. As Young Lose Interest in Cars, G.M. Turns to MTV for Help. 22 March 2012. New York Times, New York City.
- Delbosc, A., Currie, G., 2013. Causes of youth licensing decline: a synthesis of evidence. *Transp. Rev.* 33 (3), 271–290.
- Delbosc, A., Currie, G., 2013. Changing demographics and young adult driver license decline in Melbourne, Australia (1994–2009). *Transportation*, <http://dx.doi.org/10.1007/s11116-013-9496-z>. (in press).
- Department for Environment Food and Rural Affairs, 2009. Survey of Public Attitudes and Behaviours Toward the Environment: Data Tables. TNS, London.
- Forward, S., Åretun, A., Engström, I., Nölén, S., Börjesson, J., 2010. Young People's Attitudes Towards Acquiring a Driving License 2002–2009. VTI (in Swedish).
- Fox, F.E., Morris, M., Rumsey, N., 2007. Doing synchronous online focus groups with young people: methodological reflections. *Qual. Health Res.* 17 (4), 539–547.
- Golob, T.F., Regan, A.C., 2001. Impacts of information technology on personal travel and commercial vehicle operations: research challenges and opportunities. *Transp. Res. Part C: Emerg. Technol.* 9 (2), 87–121.
- Hiscock, R., Macintyre, S., Kearns, A., Ellaway, A., 2002. Means of transport and ontological security: do cars provide psycho-social benefits to their users? *Transp. Res. Part D* 7, 119–135.
- Kalmbach, R., Bernhart, W., Kleimann, P.G., Hoffman, M., 2011. Automotive Landscape 2025: Opportunities and Challenges Ahead. Roland Berger.
- KRC Research, 2010. Millennials & Driving: A Survey Commissioned by Zipcar.
- Kuhnimhof, T., Armoogum, J., Buehler, R., Dargay, J., Denstadli, J.M., Yamamoto, T., 2012a. Men shape a downward trend in car use among young adults – evidence from six industrialized countries. *Transp. Rev.* 32 (6), 761–779.
- Kuhnimhof, T., Buehler, R., Dargay, J., 2011. A new generation: travel trends for young Germans and Britons. *Transp. Res. Rec.* 2230, 58–67.
- Kuhnimhof, T., Buehler, R., Wirtz, M., Kalinowska, D., 2012b. Travel trends among young adults in Germany: increasing multimodality and declining car use for men. *J. Transp. Geogr.* 24, 443–450.
- Kuhnimhof, T., Wirtz, M., Manz, W., 2012c. Lower incomes, more students, decrease of car travel by men, more multimodality: decomposing young Germans' altered car use patterns. In: Proceedings of the Transportation Research Board 91st Annual Meeting. Washington, DC.
- Le Vine, S., Latinopoulos, C., Polak, J., 2013. A tenuous result: re-analysis of the link between internet-usage and young adults' driving-licence holding. Comments on 'Recent changes in the age composition of drivers in 15 countries'. *Traffic Injury Prev.* 14 (6), 654–657.
- Mann, C., Stewart, F., 2000. *Internet Communication and Qualitative Research: A Handbook for Researching Online*. Sage, London.
- McDonald, N., Trowbridge, M., 2009. Does the built environment affect when American teens become drivers? Evidence from the 2001 national household travel survey. *J. Saf. Res.* 40, 177–183.
- Nantulya, V.M., Sleet, D.A., Reich, M.R., Rosenberg, M., Peden, M., Waxweiler, R., 2003. Introduction: the global challenge of road traffic injuries. *Injury Control Saf. Promot.* 10 (1–2), 3–7.
- Raimond, T. and Milthorpe, F., 2010. Why are young people driving less? Trends in license-holding and travel behaviour. In: Proceedings of the Australasian Transport Research Forum. Canberra, Australia.
- Redshaw, S., 2006. Driving cultures: cars, young people and cultural research. *Cult. Stud. Rev.* 12 (2), 74–89.
- Sandqvist, K. and Kriström, S., 2001. Getting Along Without a Family Car: The Role of the Automobile in Adolescents' Experiences and Attitudes. Stockholm Institute of Education, Stockholm.
- Scheiner, J., Holz-Rau, C., 2007. Travel mode choice: affected by objective or subjective determinants? *Transportation* 34 (4), 487–511.
- Sivak, M., Schoettle, B., 2012a. Recent changes in the age composition of drivers in 15 countries. *Traffic Injury Prev.* 13, 126–132.
- Sivak, M., Schoettle, B., 2012b. Update: percentage of young persons with a driver's license continues to drop. *Traffic Injury Prev.* 13, 341.
- Sivak, M., Schoettle, B., 2013. Toward understanding the recent large reductions in the proportion of young persons with a driver's license: a response to Le Vine et al. *Traffic Injury Prev.* 16 (4), 658–659.
- Smart, M., 2013. Trip-making and the activity participation of youth: trends from 1990–2009. In: Proceedings of the Transportation Research Board 92nd Annual Meeting. Washington, DC.
- Steg, L., 2005. Car use: lust and must. Instrumental, symbolic and affective motives for car use. *Transp. Res. Part A* 39, 147–162.
- Stokes, G., Hallett, S., 1992. The role of advertising and the car. *Transp. Rev.* 12 (2), 171–183.
- VicRoads, 2012. VicRoads License Breakdown by Age, 2001–2011.
- Williams, A.F., 2011. Teenagers' licensing decisions and their views of licensing policies: a national survey. *Traffic Injury Prev.* 12 (4), 312–319.